

3D Motion Planning for UAVs in GPS-Denied Unknown Forest Environment

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Abstract—In this paper, a decomposition hierarchic on-line motion planning approach consisting of path planning and trajectory generation is proposed for VTOL UAVs to fly in a GPS-denied unknown obstacle-rich environment such as forest and urban canyon. A closed-loop 3D path planning based on A^* search algorithm is used to generate collision-free path and a 3D on-line trajectory generation based on maneuver automaton methods is used to generate a collision-free reference trajectory. The simulation and experiment on a VTOL UAV demonstrate the effectiveness of the proposed motion planning approach.

I. INTRODUCTION

Autonomous navigation of UAVs in complex environments has been an active research topic for a decade. The motion planning problem for UAVs is difficult since these vehicles have fast, complicated and uncertain dynamics [1] and strict payload limitations [2], they are compounded by the issues of real-time navigation in 3D space and involve continual interaction with the environment.

The goal of motion planning of UAVs is to generate a collision-free trajectory in real-time for kinematics/dynamics-constrained vehicles to go through a three-dimensional environment with obstacles. Currently, there does not exist an algorithm that provides an exact analytic solution to such a problem. Indeed, even state-of-the-art approximation algorithms operating on a three-dimensional subspace of this problem space are difficult to compute in real time. Many of the algorithms designed to solve this problem rely on a decomposition hierarchic method, first solving a path planning problem, applying smoothing constraints, forming a trajectory that conforms to the path, and using a control loop to follow this trajectory [3]. The advantage of this method is that any of the traditional kinematics/dynamics-unconstrained algorithms found in the field of robotics may be used for the path planning stage, and it is easy to integrate the motion planning algorithm with existing low level dynamic controller.

The algorithms for path planning are evaluated in terms of optimality, completeness and computational complexity. As it is difficult to optimize all these elements in the

development of path planning algorithm, there is always a trade-off between these elements. The existing path planning algorithms include roadmap approaches that consist of map discretization and search algorithm [4], [5], [6], [7], [8], heuristic approaches [9], [10], rapidly exploring random tree [11], [12] and artificial potential field approaches [13], [14].

Trajectory generation problems are often considered as constrained optimization problems. It solves the trajectory that is physically possible to be tracked by an UAV from one node to another subjected to dynamic constraints. An exact solution is generally not possible in three-dimensional space. As a result, approximation solutions are usually used to find a feasible or sub-optimal trajectory subject to dynamic constraints. The existing methods can be classified as smoothing methods [15], mathematical programming methods [16], [17], [18], [19], potential-based methods [20], [21] and maneuver automaton methods [22], [23], [24], etc. The general idea of maneuver automaton methods [22], [23], [24] is to reduce the optimization or search problem from an infinite-dimensional space to a finite one. It helps to significantly reduce the computational complexity of a trajectory optimization. The commercial trajectory generation software Reflexxes is based on this type of methods.

Since the computational time of deterministic and complete algorithms grows exponentially with the dimension of the configuration space, most of the existing algorithms do not provide an adequate solution for on-line (real-time) UAV motion planning in unknown natural environments. Majority of the existing motion planning methods only provide simulation results based on idealized vehicles and operational conditions and do not discuss practical implementation.

In this paper, a 3D motion planning method for VTOL UAVs to fly in a GPS-denied unknown obstacle-rich environment such as forest and urban canyon is proposed. A closed-loop on-line path planning based on A^* search algorithm is used to generate collision free path with the shortest distance from any current position to a given destination in the local map of the UAV. Then an on-line trajectory generation based on maneuver automaton methods generates a collision-free reference trajectory and output to the lower level of control system to control the VTOL UAV. This proposed motion planning method has some inherent, implicit robustness to uncertainties due to its closed-loop implementation.

Since we utilize existing techniques for sensing, map building, path planning and trajectory generation, our contribution is not so much to these individual fields, but rather one of combining these techniques in a novel way to facilitate 3D navigation of a UAV in GPS-denied unknown environments.

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This paper describes how we integrate these techniques, and presents insights learned in the process. Both simulation and flight test results are presented in this paper.

The rest of this paper is organized as follows. Section II presents the development of 3D motion planning algorithm. Section III gives the simulation results of the proposed algorithm. The experimental results are presented in Section IV, followed by a brief conclusion in Section V.

II. 3D MOTION PLANNING OF UAV IN UNKNOWN ENVIRONMENT

Motion planning of UAVs in unknown environment is faced with a large amount of uncertainties such as environment uncertainty, measurement noise, sensor limitation and wind disturbance, etc. Moreover, collision with obstacles is catastrophic for UAVs. Hence, soundness is a very crucial property for motion planning of UAVs. A sound motion planner is one that always guarantees the vehicle will enter the goal region and stop there without hitting any obstacle despite uncertainty in sensing and control.

As the majority of UAVs currently in production already have a low-level tracking control system and the decomposition hierarchic motion planning method is the easiest one to integrate with the existing control scheme. In this paper, the decomposition hierarchic method is adopted to implement motion planning of UAVs in unknown obstacle-rich environment. This method consists of an on-line path planning that generates a collision-free path and an on-line trajectory generation that generates a collision-free trajectory. Both of them work together to achieve the soundness of the proposed motion planning. The diagram of motion planning of a single UAV is shown in Figure 1.

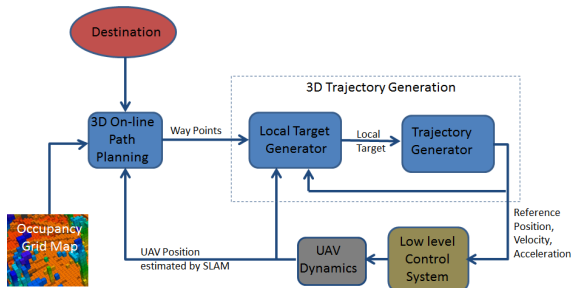


Fig. 1: Diagram of motion planning of single UAV

A. Environment Representation

In this research, the environment is detected by LIDAR sensors. The point clouds generated by the LIDAR sensors are used to build the 3D occupancy grid map surrounding the UAV. 3D occupancy grid map represents the 3D environment by dividing the environment into evenly cubes and labelling each cube as free (w/o obstacle) or occupied (with obstacle). The resolution of the occupancy grid map is a trade-off result after considering the accuracy of the environment representation, the physical size of the UAV and on-line implementation.

As the environment is unknown and the computational time of motion planning algorithms grows with the size of the environment, to implement on-line motion planning, a local but not global occupancy grid map is built based on the sensed data. In the local occupancy grid map, a fixed area surrounding the UAV is represented by its corresponding occupancy grid map and the rest area is represented by free spaces. The size of the fixed area depends on the sensing range of LIDAR and the fixed area moves with the UAV. Figure 2 shows the local occupancy grid map on which the 3D A^* algorithm works.

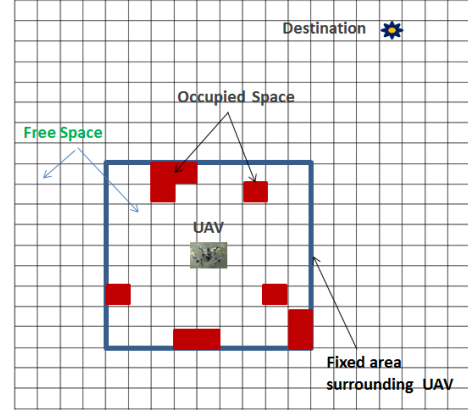


Fig. 2: Local occupancy grid map.

To make the UAV keep a safe distance with the obstacles, each occupied cube in the local map is expanded, i.e., the neighbors of the occupied cube are all set as "occupied".

B. 3D Path Planning

Path planning is referred to the generation of a feasible flight path from a start position to a destination without consideration of dynamic feasibility.

As A^* algorithm is a simple, fast, complete and optimal algorithm, we use A^* to search the shortest path from the start node to the destination node in the 3D local occupancy grid map. While A^* traverses the map, it builds up a tree of partial paths. The leaves of this tree (called the open set) are stored in a priority queue that orders the leaf nodes by a cost function. The cost function $f(n)$ is described by

$$f(n) = g(n) + h(n) \quad (1)$$

where $g(n)$ is the distance travelled from the initial node to node n and its value is tracked by the algorithm, $h(n)$ is a heuristic estimate of the cost to reach the goal node from node n . For the algorithm to find the actual shortest path, the heuristic function must be admissible, namely, it never overestimates the actual cost to get to the destination node. One candidate for $h(n)$ is the Euclidean distance from node n to the destination node as it is physically the smallest possible distance between any two points. In the following, the 3D A^* algorithm developed in this research is given:

Step 1: Initialization: Define the start node as node s . Let $g(s) = 0$ and the cost function $f(s) = g(s) + h(s)$ with $h(s)$ the Euclidean distance from the start node s to the destination

node t . Set the closed set consisting of all occupied nodes (obstacles) in the occupancy grid map and node s . And set node s as its parent node.

Step 2: Node Expansion: Find free neighbour nodes of the node n ($n = s$ at the start node) from its 26 neighbour nodes. Save the cost and corresponding parent of these free neighbour nodes. Keep these free neighbour nodes in an open set in the order of cost value.

Step 3: Node Selection: If there is no node in the open set, display "No Path" and exit from this algorithm; otherwise, identify the node $n + 1$ with minimum cost in the open set. Delete this node from the open set and put it into the closed set, and save its parent node.

Step 4: Stopping Criteria: If the node $n + 1$ is the destination, go to step 5; otherwise, let $n = n + 1$ and go to step 2.

Step 5: Short Cut: If there is no obstacle on the straight line between a node i (start from the destination node t) and its parent's parent, replace its parent with its parent's parent. Otherwise, let $i = i + 1$ and repeat Step 5 until the start node is reached.

Step 6: Path Finding: From destination node and its parent and its parent's parent, and so on, by using backward induction, the shortest path from start node s to destination node t can be found.

It is noted that the path generated by the A^* algorithm may not be the shortest path on the global map since the A^* algorithm works on the local occupancy grid map. However, the computational time will be reduced and the completeness of the A^* algorithm will not be affected.

C. 3D Trajectory Generation

As the path generated by the on-line path planning does not consider kinematic/dynamic feasibility, it is difficult for a UAV to follow it without stopping. As a result, on-line trajectory generation is required to connect the high-level path planning with the low-level motion control of the UAV such as the low-level motion control can control the UAV with kinematic/dynamic constraints flying along the path generated by the high-level path planning.

In this paper, Reflexxes Type IV Motion Library [25] based on maneuver automaton methods is adopted to generate 3D trajectory for UAVs. This trajectory generator is capable of computing an entire time-optimal trajectory from an arbitrary initial state of motion to a given target within one servo control cycle (typically one millisecond or less) subject to velocity, acceleration and jerk limitations. Figure 3 shows the inputs and outputs of the position-based Reflexxes trajectory generator. As Reflexxes trajectory generator directly uses its outputs as its inputs of current state of motion for the following control cycle, it is considered as an open-loop control.

Since only kinetic time-optimality is regarded in this on-line trajectory generation algorithm, all DOFs can be considered as linearly independent, that is, decoupled. For multiple DOF systems, one more step is to synchronize all

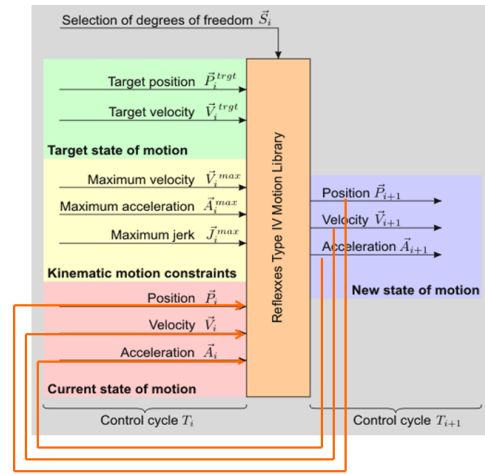


Fig. 3: Inputs and outputs of the on-line trajectory generator [25].

DOFs to the synchronization time and calculate all trajectory parameters based on the synchronization time.

The main reason to apply this algorithm to the research is that it generates 3D time optimization trajectory with consideration of kinetic constraints and can be implemented in real time. However, its drawbacks are 1) only symmetrical velocity, acceleration and jerk constraints are considered in this algorithm and UAV dynamic constraints are not addressed. 2) obstacle avoidance is not considered.

As VTOL UAVs but not fixed-wing UAVs are used in this research, the first drawback is not so significant. However, the second drawback must be solved here.

Since Reflexxes trajectory generator is open-loop and does not consider obstacle avoidance, the generated trajectory may cause the UAV colliding with obstacles. Hence, a collision check on the trajectory is necessary to ensure soundness. As a result, an intermediate element between the path planning and Reflexxes trajectory generator, called local target generator, is introduced to carry out the collision check of the trajectory.

The aim of local target generator is to generate a local target for Reflexxes trajectory generator such that the trajectory generated by Reflexxes trajectory generator from the current position to the local target is collision-free.

Since the path generated by the 3D path planning algorithm consists of a serial waypoints arranged in a pre-defined order, the local target generation algorithm selects the waypoints one by one as the local target input to Reflexxes trajectory generator. To switch the local target from one waypoint to the next one, collision check of the trajectory from current position to the next waypoint generated by Reflexxes is carried out. If it is obstacle-free, the local target switches to the next waypoint; otherwise, the local target keeps unchanged. The flow chart of the local target generation is given in Figure 4. In the local target generation, collision checking of trajectory is the most important task. The following steps are used to carry it out:

Step 1: Reconstruct the reference trajectory using the variables provided by Reflexxes. The reference trajectory

is a 3-order polynomial function of time in each degree of freedom.

Step 2: Discrete the reference trajectory to obtain a serial of discrete-time points.

Step 3: Check if the cubes where these discrete points lie are obstacle-free in the local occupancy grid map. If all the cubes are obstacle-free, it indicates that the reference trajectory is collision-free; otherwise, it indicates that there is obstacle on the reference trajectory.

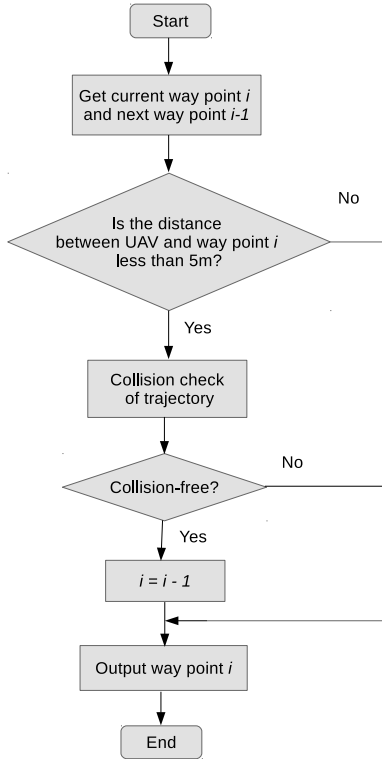


Fig. 4: Flow chart of the local target generation algorithm.

As the update rate of the 3D on-line path planner relies on the update rate of the 3D occupancy grid map which requires long processing time and the 3D Reflexxes trajectory generator has to provide commands for low-level motion control of UAVs in real-time, different time scales are required for the path planning and trajectory generator. In this research, the update rate of the path planning and local target generator are set as 10Hz and that of Reflexxes trajectory generator is set as 50Hz. All the software development including building of occupancy grid map, SLAM, 3D path planning and 3D on-line trajectory generation is under Robot Operating System (ROS).

III. SIMULATION OF MOTION PLANNING

In this section, a VTOL UAV is used to demonstrate the proposed motion planning algorithm in MatLab.

The simulated environment is a forest. The constraints of the VTOL UAV are as follows: the maximum velocity $v_{xmax} = v_{ymax} = v_{zmax} = 1m/s$, the maximum acceleration $a_{xmax} = a_{ymax} = a_{zmax} = 0.5m/s^2$ and the maximum jerk $j_{xmax} = j_{ymax} = j_{zmax} = 1m/s^3$. The

start position is set at (25,5,1) and the target position is set at (45,57,2.5).

Figures 5 and 6 show the reference trajectory (in green line) and the real trajectory (in black line) of the UAV from 3D view and X-Y view, respectively, where the red cubes represent the obstacles. The path generated by the 3D on-line path planner is shown in blue line where the blue circles are the way points that construct the path. It is observed from Figures 5 and 6 that the path generated by the 3D on-line path planner is an obstacle-free path. However, it is difficult for UAV to follow without stop at the way points as the path consists of a serial of straight lines. It is also observed that the reference trajectory (in green line) generated by the 3D trajectory generation subject to the kinetic constraints is along the path (in blue line) and is collision-free. Moreover, the real trajectory (in black line) of the UAV tracks the reference trajectory well. Moreover, the reference velocity and acceleration generated by the proposed algorithm and the real velocity and acceleration of the UAV are shown in Figure 7. This simulation demonstrates that the 3D motion planning works well in the obstacle-rich 3D environment.

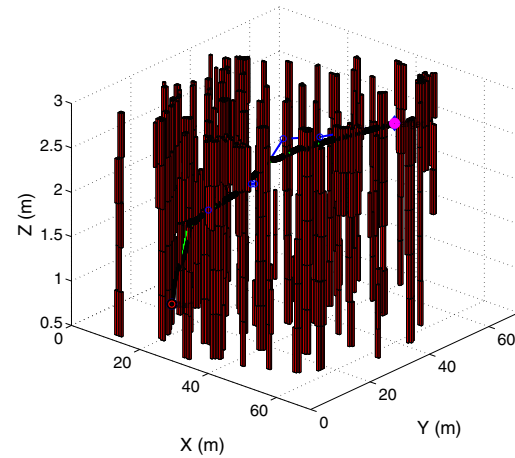


Fig. 5: 3D view of trajectory of 3D motion planning

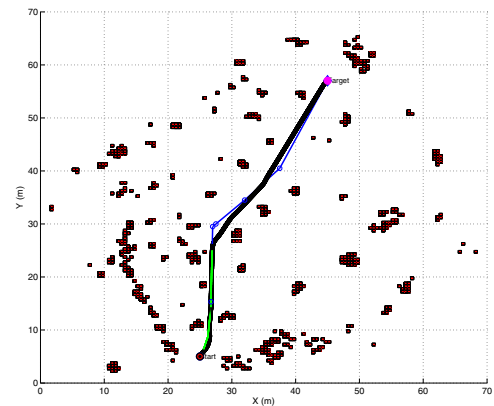


Fig. 6: X-Y view of trajectory of 3D motion planning

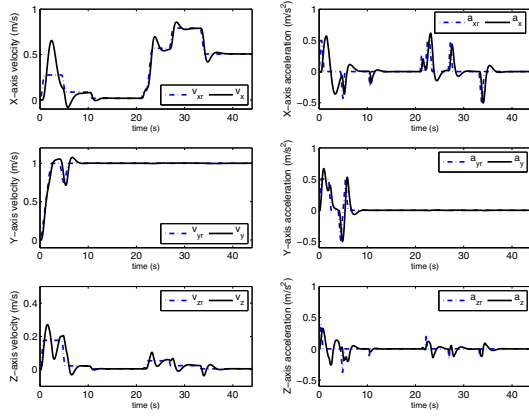


Fig. 7: Velocity and acceleration of 3D motion planning



Fig. 8: UAV for flight test.

IV. FLIGHT TEST OF MOTION PLANNING

The VTOL UAV used for flight test is shown in Figure 8. It is with the three dimension size of $1.2m \times 1.2m \times 0.5m$. This quadrotor UAV is equipped with two LIDAR scanners to detect obstacles in horizontal and vertical direction, one ranger finder to measure the relative altitude, IMU to measure Euler angles, angular rates and acceleration on the body axis, Pixhawk to perform the low-level control, and on-board processor Mastermind to perform high level control such as path planning, trajectory generation and logic algorithms, etc.

The flight test was carried out in a forest. Part of the occupancy grid map representing the forest is shown in Figure 9 where obstacle-occupied space is represented by colourful pixels and free space by transparent pixels.

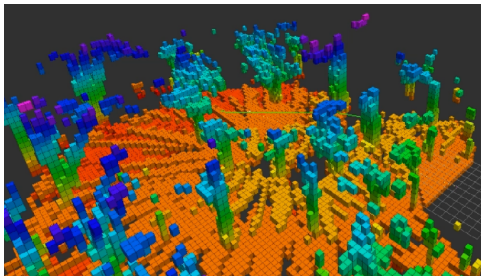


Fig. 9: Occupancy grid map of the forest environment

In the flight test, the UAV flew a loop from the start position $(0,0,1.2)$ to the destination at the same position

$(0,0,1.2)$ via a serial given middle points. The reference trajectory generated by the motion planning and the trajectory estimated by SLAM are shown in Figure 10 where the red squares represent the obstacles, the blue stars represent the given middle points, the black line represents the trajectory estimated by SLAM and the green line represents the trajectory generated by the proposed motion planning algorithm. It is observed from Figure 10 that the reference trajectory generated by the proposed approach is collision-free and the UAV tracks the reference trajectory very well in the flight test.

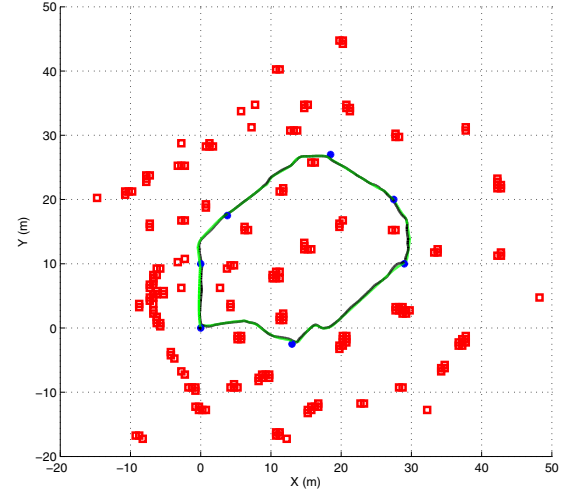


Fig. 10: XY-view of trajectory in flight test where the red squares represent the obstacles, the blue * represent the middle points, the black line represents the trajectory estimated by SLAM and the green line represents the reference trajectory generated by the proposed approach.

Figures 11 shows that the reference velocities generated by the motion planning is tracked by the UAV well. In addition, the acceleration of the UAV is shown in Figure 12 and the Euler angles and angular rates are shown in Figure 13.

The flight tests demonstrate that the proposed motion planning algorithm works effectively in the obstacle-rich forest environment.

V. CONCLUSIONS

In this paper, a decomposition hierarchic on-line motion planning method is proposed for 3D motion planning of UAVs in obstacle-rich forest environment. Since the proposed motion planning forms a closed-loop with low level control system by introducing the UAV position feedback, it has some inherent, implicit robustness to uncertainties. The simulation and experiment on a single UAV demonstrate the effectiveness of the proposed on-line 3D motion planning algorithm.

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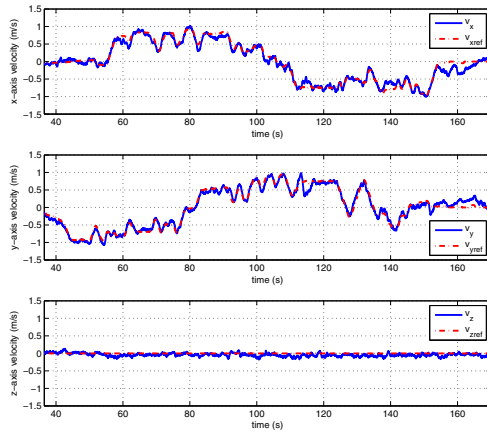


Fig. 11: Velocities in flight test.

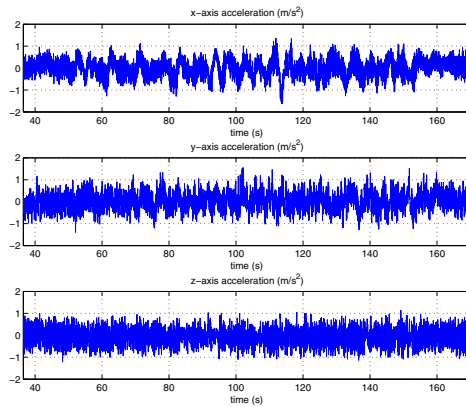


Fig. 12: Accelerations in flight test.

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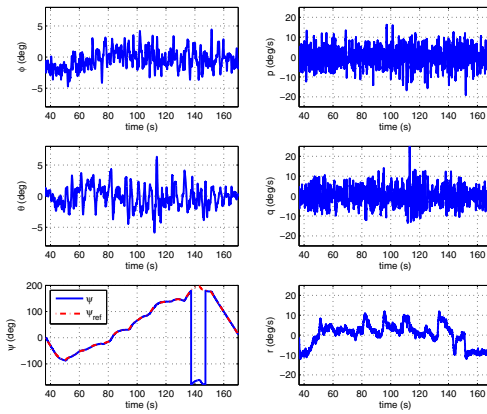


Fig. 13: Euler angles and angular rates in flight test.

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